

Frozen tuna is unloaded from a refrigerated ship in Japan, 2005



FROZEN CARGO

Refrigeration not only allowed food to be kept longer at home, it has also changed what we eat. The invention of refrigerated ships in the 1870s transformed the global food supply. Easily spoiled items, such as meat and fish, can now be frozen for transporting across the world.

FLASH FREEZING

The American naturalist Clarence Birdseye developed the rapid refrigeration method known as flash freezing in the 1920s, based on a technique used by the Inuit people of Canada. It involves cooling food quickly at very low temperatures, which keeps it fresher than if it is frozen more slowly.



Workers using flash freezing to preserve seafood

AT HOME



COOLER INDOORS

Invented in 1902 by the American engineer Willis Carrier, air conditioning works much like refrigeration. A coolant loops between the inside of a building, where it absorbs heat, and the outside, where the heat is released into the air with the help of fan units, as shown in the picture above.

SMART REFRIGERATORS

Since the late 1990s, many companies have tried to perfect the smart refrigerator—a device that knows when it's running low on a product and orders new supplies automatically via the Internet. Samsung's Family Hub Refrigerator (below) has a camera inside that lets owners check the contents of their fridge using a smartphone, even if they are not at home.

